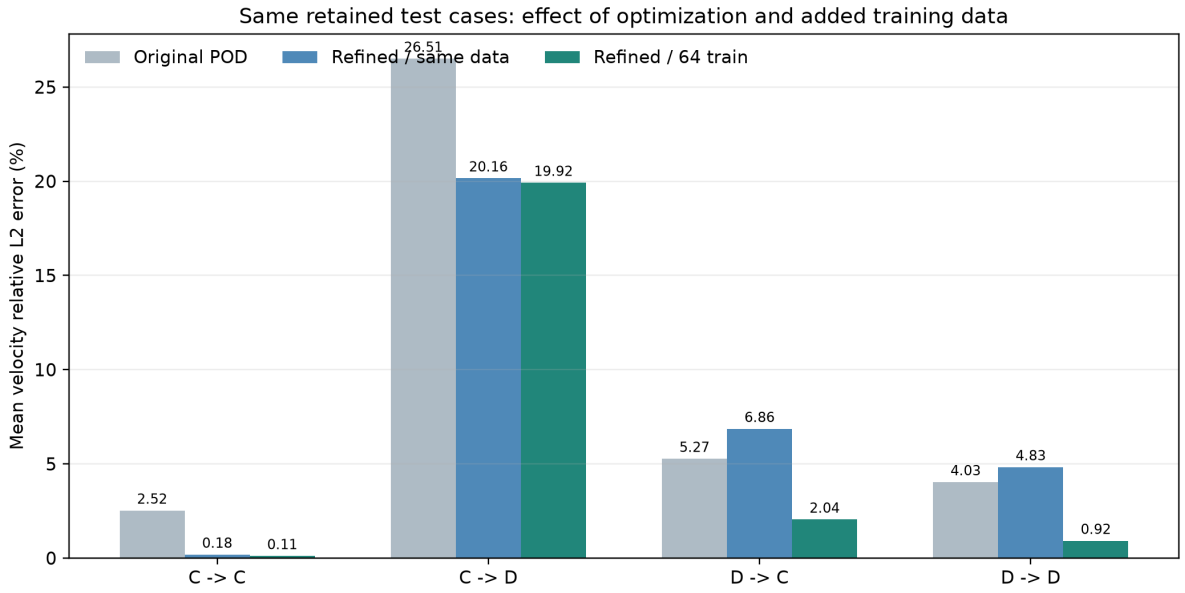
A physical low-fidelity field becomes the coordinate for a nonlinear correction.

The network receives an actual Stokes solution at the same geometry and lid boundary condition, plus Reynolds number. It predicts a correction to streamfunction; velocity follows by differentiation and pressure is recovered from momentum afterwards.



0.107%	0.917%	2.044%	19.920%
constant -> constant	diverse -> diverse	diverse -> constant	constant -> diverse

Evidence boundary. *These are mean velocity relative L2 errors on unchanged, previously inspected n=25 regression tests. They demonstrate surrogate improvement against same-grid numerical labels, not mesh-independent CFD accuracy or a new neural architecture.*

Stokes	MLP	Velocity	Pressure
matched low-fidelity field	POD correction coefficients	derivatives of corrected psi	momentum recovery

Physical formulation

1. Steady incompressible flow

$$div(u) = 0, \quad (u \cdot grad)u = -grad(p) + Re^{-1} Laplacian(u)$$

With streamfunction psi, u = d psi/dy and v = -d psi/dx, so incompressibility is satisfied by construction. The Navier-Stokes labels retain convection; the Stokes baseline removes only convection while keeping the same cavity, grid, and lid data.

2. Reynolds number changes with speed

$$Re = U_{ref} L / nu, \quad U_{ref} = Re\ nu / L, \quad L = 1, \quad nu = 0.0025$$

Changing lid speed therefore changes Reynolds number. A dimensionless Stokes velocity field by itself does not identify the nonlinear target; Re is supplied explicitly.

3. Learned correction

$$psi_{NS} = psi_S + Delta\ psi_{theta}(psi_S, Re)$$

The prediction target is the streamfunction difference. Velocity is obtained from derivatives of the corrected streamfunction. Pressure is recovered through a least-squares momentum-gradient solve and is never a network output.

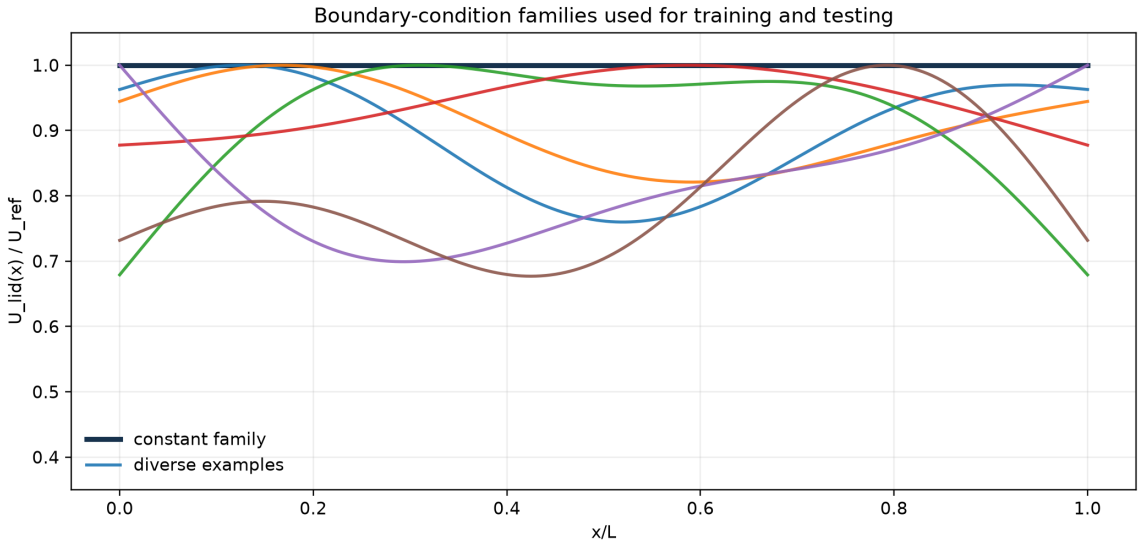
> *Why this is useful: Stokes supplies geometry and boundary-condition response exactly for the linear problem; learning is reserved for the Reynolds-dependent nonlinear departure.*

Quantity	Stokes baseline	Learned correction	Final field
Convection	omitted	learned implicitly	represented
Boundary data	exact input	not relearned	matched
Incompressibility	streamfunction	streamfunction	by construction
Pressure	not used as target	not predicted	recovered afterwards

Dataset design: constant and diverse lids

The two training families use identical Reynolds-number distributions. This isolates boundary-condition diversity rather than confounding it with a different Re range.

$$\text{Constant: } U_{lid}(x) = U_{ref}$$
$$\text{Diverse: } U_{lid}(x) = U_{ref}[1 + \text{sum } a_k \sin(k\,\pi\, x/L)] \,/\, \max |1 + \text{sum } a_k \sin(k\,\pi\, xi/L)|$$



Group	Train	Validation	Test	Purpose
constant	64	16	6	Uniform lid
diverse	64	16	6	Three-mode positive lids
shape OOD	0	0	6	Five modes, larger amplitude
Re OOD	0	0	6	450 < Re < 600

Original test cases are frozen across the old, same-data refined, and expanded comparisons. Added cases enter training and validation only.

Numerical solvers and data isolation

Navier-Stokes labels

Streamfunction-vorticity finite differences; midpoint RK2; discrete-sine-transform Poisson solve; Thom wall-vorticity closure; n=25; steady PDE residual checked twice before a field becomes a label.

Stokes input

One sparse block finite-difference system with convection removed; same n=25 grid, lid samples, streamfunction convention, and wall closure; LU factorization reused across cases.

1 **Generate complete CFD cases**

No spatial patches are split across train and test.

2 **Solve matched Stokes fields**

No Navier-Stokes target enters the input solver.

3 **Fit POD bases on training only**

Validation selects rank/width; test remains outside selection.

4 **Average three selected seeds**

Seeds 7, 17, and 27; ensemble rule fixed before test reporting.

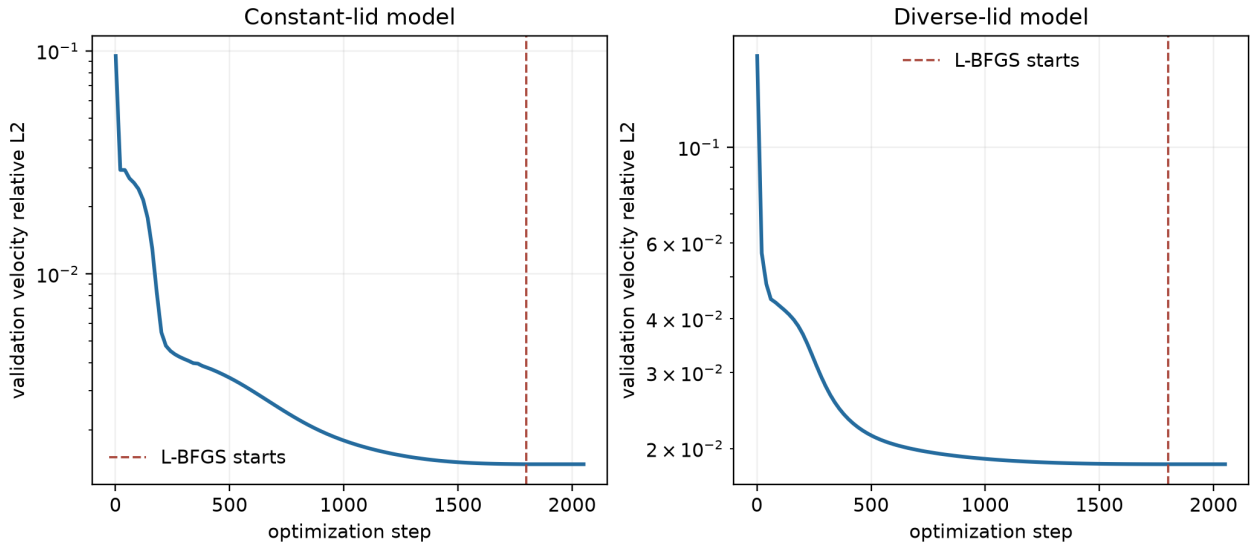
Retained numerical audit	Observed value	Interpretation
Manufactured psi error, n=17/33/65	0.0378/0.0095/0.0024	decreases by about 4x
Observed refinement order	1.995/1.999	second order
Linear relative residual	2.62e-16	sparse solve passed
One cached Stokes solve	0.282 ms	single CPU case
Unmodified Stokes velocity error	32.45%	correction is necessary

POD correction network and optimization

Architecture

Training-only POD compresses the Stokes field. A second training-only POD represents the correction. The selected MLP receives [Re/400, log(Re/400), normalized Stokes POD coefficients] and uses two 96-neuron tanh hidden layers with a linear rank-24 output.

$$\textit{Loss} = \textit{field error} + \textit{velocity-gradient error} + \textit{weighted vorticity error}$$



Family	Seed	Validation L2	Selected step	Fit time (s)
constant	7	0.00140	1800	2.25
constant	17	0.00153	1800	1.76
constant	27	0.00146	1800	2.33
diverse	7	0.01839	1800	2.13
diverse	17	0.01953	1800	2.37
diverse	27	0.02121	1800	2.34

Adam runs for 1800 epochs with cosine decay. L-BFGS starts from the validation-selected Adam checkpoint, not the final Adam iterate. Architecture and checkpoint selection do not inspect test errors.

Input POD

computed from training Stokes fields only

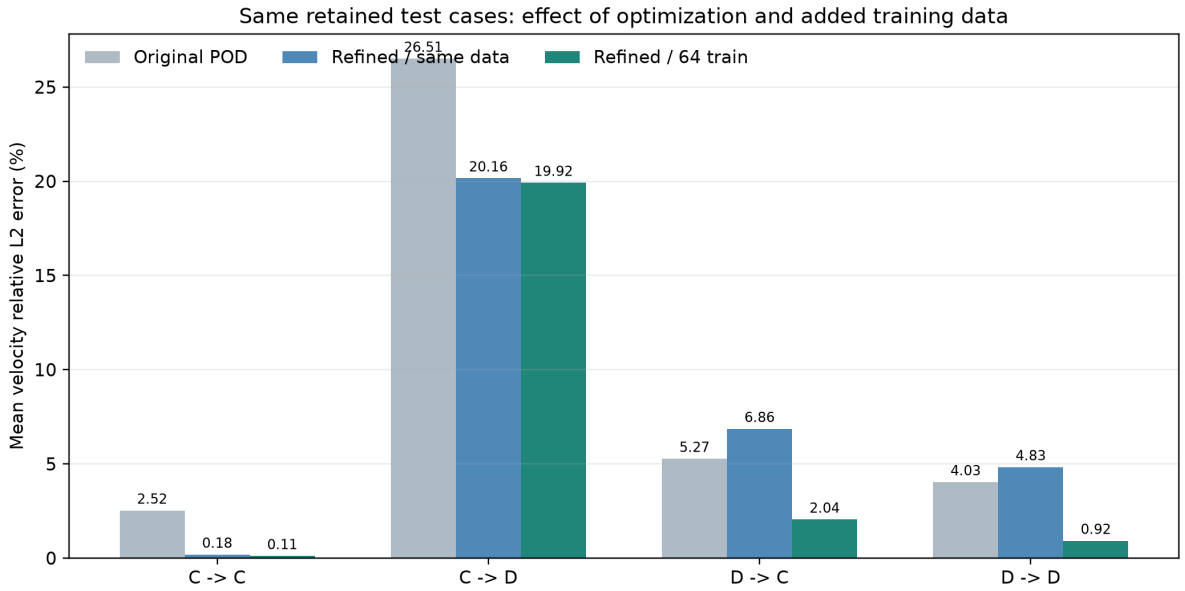
Correction POD

computed from training NS - Stokes differences

Ensemble

three independently fitted selected networks

Quantitative results and generalization



Train family	constant	diverse	shape OOD	Re OOD
constant	0.107%	19.920%	37.192%	36.982%
diverse	2.044%	0.917%	8.428%	10.204%

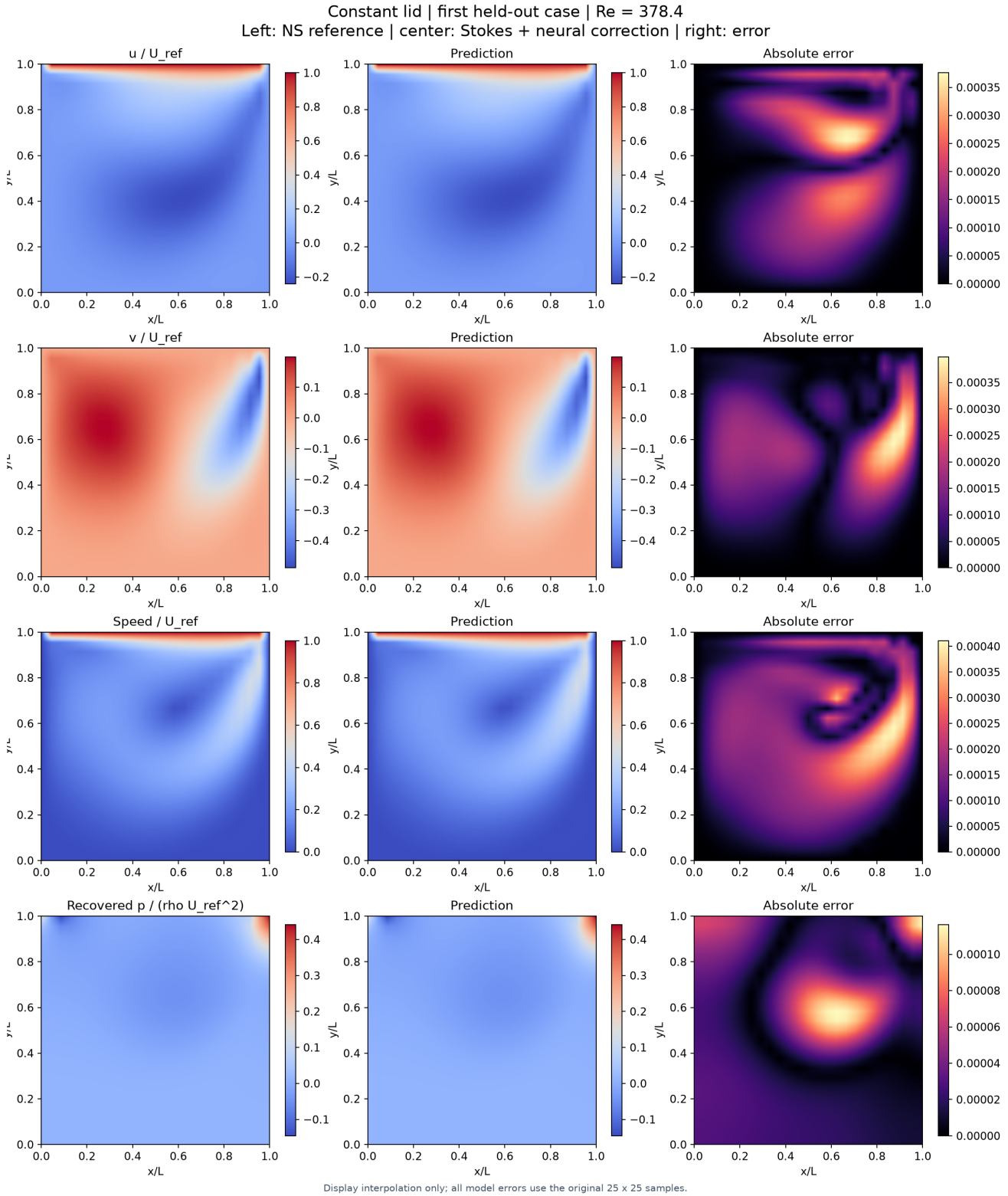
Main scientific lesson. Boundary-condition diversity matters more than simply reducing in-family error. The diverse model reaches 2.044% on constant lids, 0.917% on diverse lids, 8.428% on shape OOD, and 10.204% on Reynolds OOD. Constant-only training remains at 19.920% on diverse lids and above 36% on both OOD groups.

Pressure is a secondary consistency check. For diverse-to-diverse cases, mean full/core recovered-pressure errors are 1.12% and 2.51%. Both reference and prediction use the same recovery operator.

Worst retained case	constant	diverse	shape OOD	Re OOD
constant	0.14%	39.81%	62.14%	56.34%
diverse	3.01%	1.92%	12.86%	17.07%

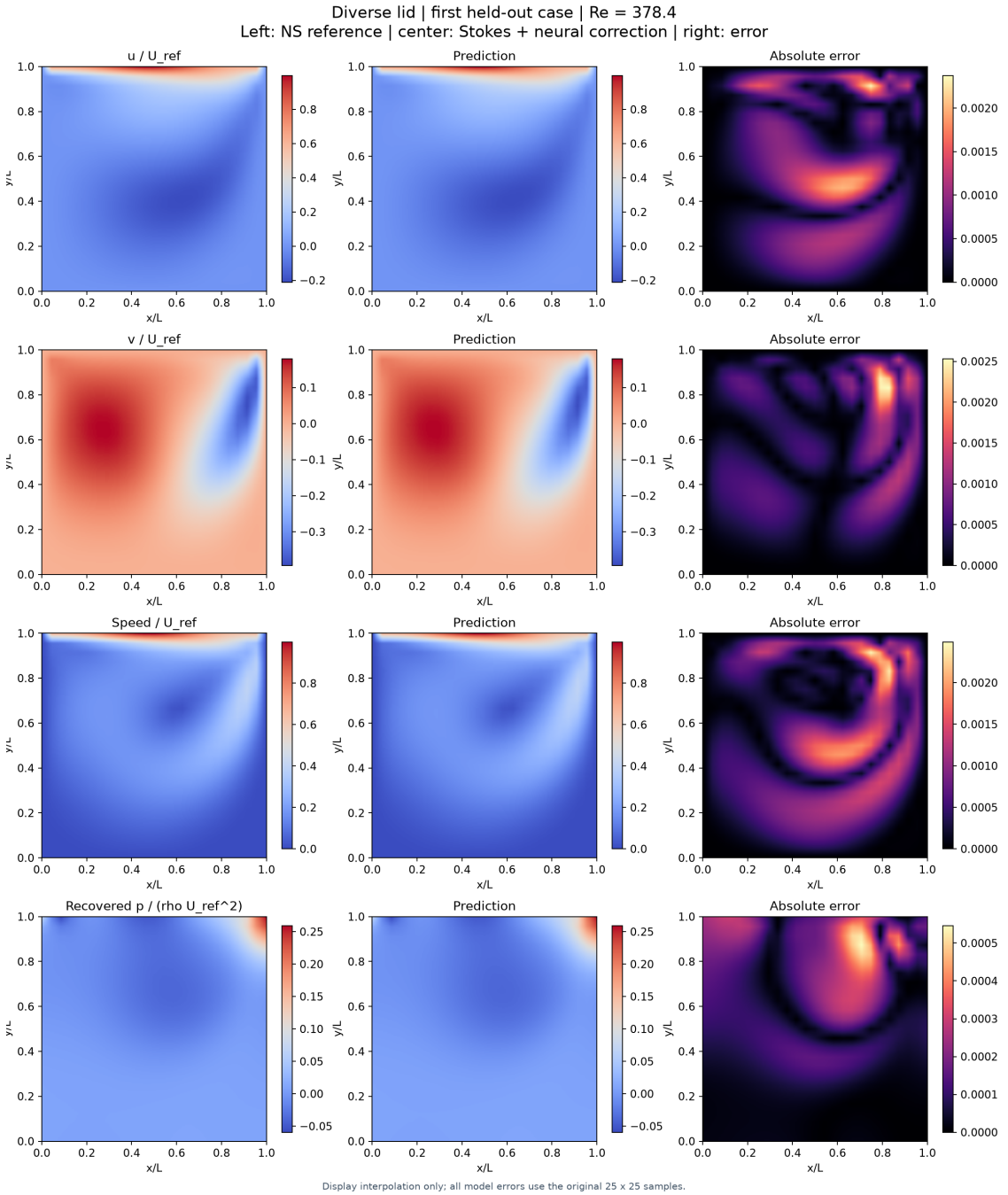
Held-out constant-lid field

First test case in manifest order; never selected by prediction quality. Left: Navier-Stokes reference. Center: Stokes plus learned correction. Right: absolute error. Shared reference/prediction scales are used within each row. Linear interpolation smooths display only; metrics use the original n=25 grid.



Held-out diverse-lid field and claim boundary

The diverse model is evaluated on the first diverse-lid test case in manifest order. The pressure row is a recovered field rather than a network output. Linear interpolation smooths display only; metrics use the original n=25 grid.



What is established

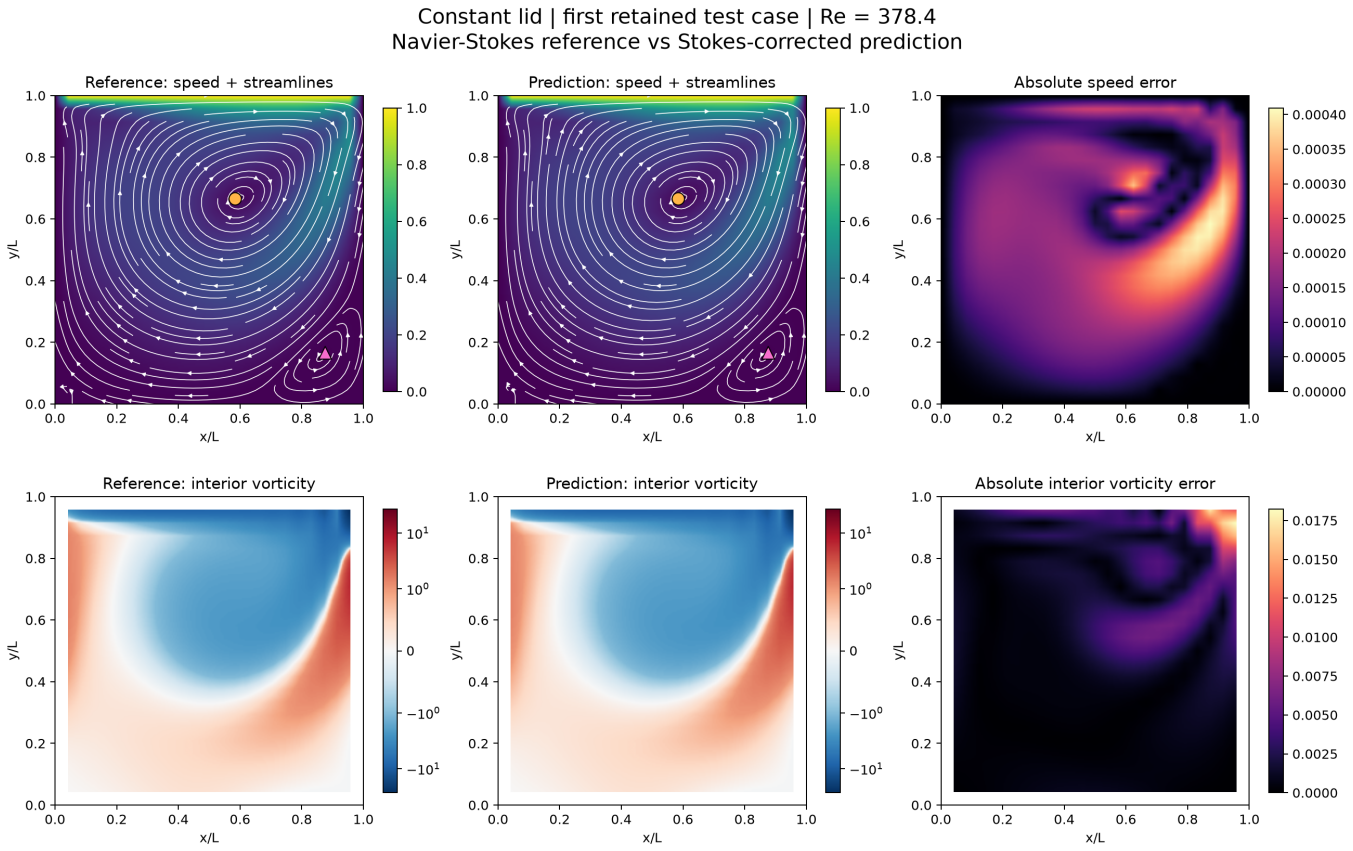
Actual Stokes input; consistent Re-speed coupling; complete-case splits; validation-only selection; three-seed ensemble; strong in-family correction; useful transfer from diverse to constant lids.

What is not established

Mesh independence; high-fidelity CFD validation; independent pressure validation; a new network architecture; universal transfer from a constant lid to unseen boundary shapes.

Constant-lid vortices and streamlines

Speed colors and white streamlines compare the same retained test case. The orange circle is the primary streamfunction minimum; the pink triangle marks a positive lower-right streamfunction maximum. Vorticity is the negative discrete Laplacian of streamfunction in both fields, with a shared symmetric-log color scale.



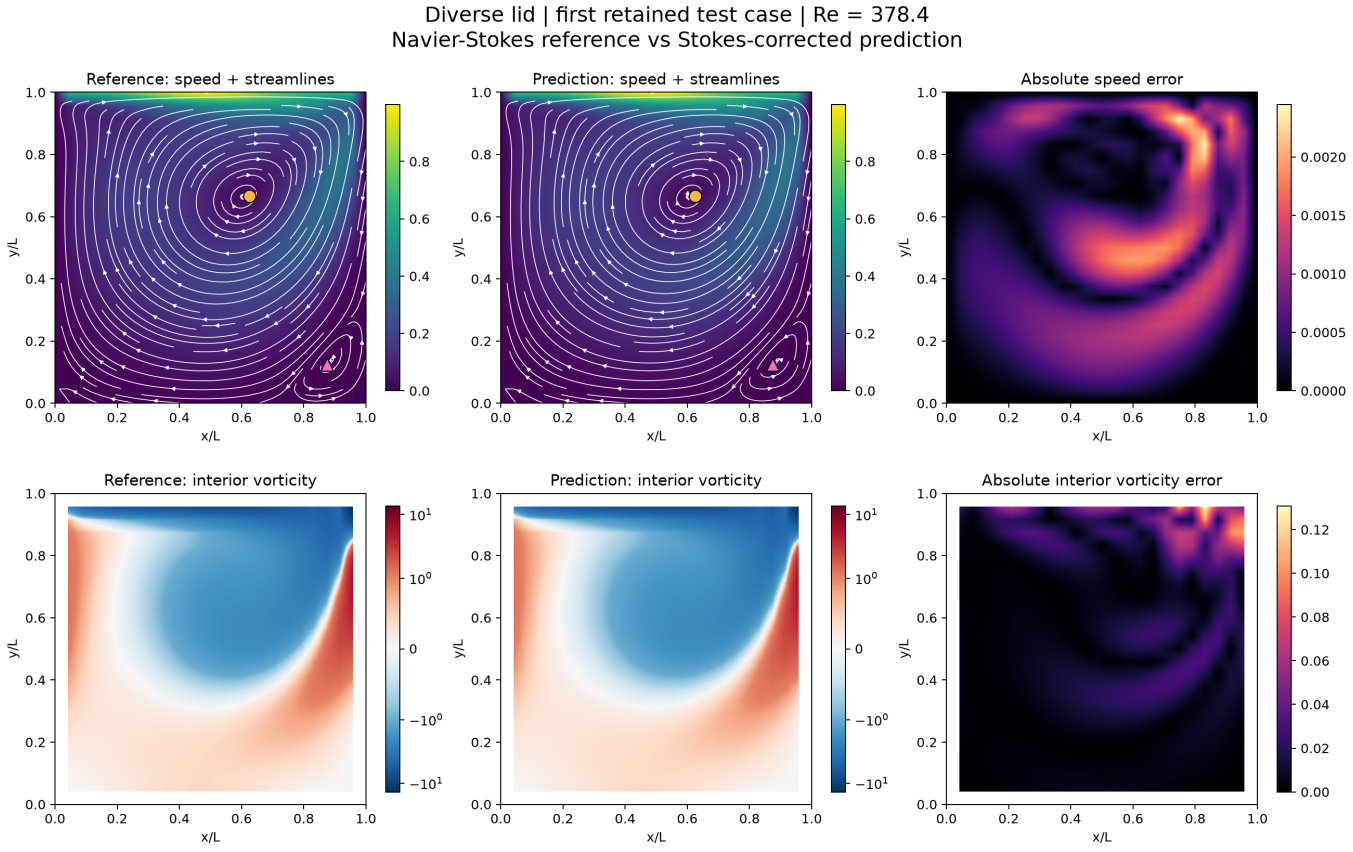
Orange circle: primary vortex; pink triangle: lower-right recirculation. Across 6 retained constant-lid tests: centers match 6/6 and 6/6 grid nodes; mean interior vorticity error 0.12%.

Six retained tests	Observed	Meaning
Primary-vortex grid node	6/6 matched	same sampled center
Lower-right local maximum	6/6 matched	small recirculation candidate
Interior vorticity relative L2	0.12% mean; 0.15% worst	same-grid comparison

Display colors are linearly interpolated for legibility. Streamlines follow the retained n=25 velocity field; vortex positions and vorticity errors are measured on original grid nodes. Agreement here does not establish mesh-independent capture of the small corner vortex.

Diverse-lid vortices and streamlines

Speed colors and white streamlines compare the same retained test case. The orange circle is the primary streamfunction minimum; the pink triangle marks a positive lower-right streamfunction maximum. Vorticity is the negative discrete Laplacian of streamfunction in both fields, with a shared symmetric-log color scale.



Six retained tests	Observed	Meaning
Primary-vortex grid node	6/6 matched	same sampled center
Lower-right local maximum	6/6 matched	small recirculation candidate
Interior vorticity relative L2	1.49% mean; 3.24% worst	same-grid comparison

Display colors are linearly interpolated for legibility. Streamlines follow the retained n=25 velocity field; vortex positions and vorticity errors are measured on original grid nodes. Agreement here does not establish mesh-independent capture of the small corner vortex.